Exploratory Data Analysis (EDA)

is a critical initial step in the data analysis process? It involves examining and understanding the data to summarize its main characteristics, often using visual methods. Here are some key aspects and techniques involved in EDA:

1. **Data Quality Checks**: Verifying data integrity, consistency, and correctness to ensure reliability in subsequent analyses.

2. **Cleaning The Data:**
   a. **Handling Missing Data**: Identifying and dealing with missing values through imputation or deletion, depending on the nature and extent of missingness.
      a. **Remove/Manipulating uneven Values:**
      b. **Replace the missing Values**
   b. **Summary Statistics**: Calculating measures like
      a. mean, : average Value
      b. median, : average Value
      c. mode, : most frequent
      d. variance,
      e. standard deviation
   c. **Outlier Detection and Removal**: Identifying data points that significantly differ from the rest of the data, which can impact the analysis and modeling process.

to describe the data numerically.

3. **Data Visualization**: Creating charts such as histograms, box plots, scatter plots, and density plots to visualize the distribution, relationships, and patterns within the data.
   a. **Identifying Trends and Patterns**: Looking for trends, seasonal variations, or cyclical patterns in time-series data or across different segments of the dataset.

   b. **Understanding Relationships**: Exploring correlations and dependencies between variables to understand how they interact and influence each other.

4. **Data Transformation**: Transforming variables (e.g., normalization, standardization) to meet the assumptions of certain statistical techniques or improve model performance.

5. **Initial Hypothesis Generation**: Formulating initial hypotheses or insights based on the observed patterns and relationships in the data.

EDA helps analysts and data scientists to gain insights into the data, understand its limitations, and prepare for further statistical modelling or machine learning tasks. It's a crucial phase that lays the foundation for more advanced analyses and decision-making processes.

EDA for Machine Learning:
1) **Stats-based Analysis (Mean, median, STD)**
2) **Graph-Based Analysis (Different type of graphs)**
3) **Pre-processing:**
   1) Handle Missing Value
   2) Handel Duplicate Values
   3) Basic Cleaning (bring data in to right values/format)
   4) Outlier Handling
   5) Handel Imbalance Data
   6) Feature Selection
   7) Dimension Reduction
   8) Encoding
   9) Scaling

Handle Missing Value:
   1) Random value
   2) Forward filling/backward filling
   3) Statistical approach (mean. mode, median)
   4) End of the distribution
   5) Create "your own" ML model to predict missing value
   6) Use mean, median, mode to replace the missing value

Handel Duplicate Values:
   1) Drop the duplicate

Handling imbalance Dataset
   1) Collect more data
   2) Under sampling
   3) Over sampling
   4) Cluster based oversampling

Feature Selection
Dimension Reduction
   1) (PCA, LDA, TSNE)
   2) Split/Merge/Drop/Add

Outlier Handling:
a) Detect the Outlier
   1) z-Score
   2) IQR
   3) Boxplot
   4) Scatter plot
   5) Violin plot

b) Removing Outlier
   1) Drop Outlier
   2) Replace with other value

Scaling:
   1) Standardization
   2) Min-max scaler

Encoding
   1) One hot encoding
   2) Ordinal Encoding